Link Layer & LANs Definitions

6.1 Introduction to the Link Layer

Node Any device in a network, including hosts and routers.
Link A communication channel connecting two adjacent nodes.
Frame The Layer-2 (link layer) packet. It encapsulates a network-layer datagram.
Link Layer Responsibility To transfer a datagram from one node to a physically adjacent node over a single link.
Link Layer Interface Communicating
- **Sending Side:** Encapsulates the datagram in a frame; adds error checking bits, reliable data transfer info, and flow control.
- **Receiving Side:** Looks for errors, reliability, etc.; extracts the datagram and passes it to the upper layer.

Services Provided
- **Framing:** Encapsulating a datagram into a frame, adding header and trailer.
- **Flow Control:** Pacing between adjacent sending and receiving nodes to prevent the sender from overwhelming the receiver.
- **Hop-by-Hop Reliable Delivery:** Ensures a frame is successfully transmitted across a single link (often used for high-error links like wireless).

Transmission Modes
- **Half-Duplex:** Nodes at both ends of the link can transmit, but *not* at the same time.
- **Full-Duplex:** Nodes at both ends of the link can transmit and receive simultaneously.

NIC (Network Interface Card) The hardware (and associated software/firmware) in a host or router that implements the link-layer and physical-layer protocols.

6.2 Error-Detection and Correction Techniques

Error Detection Identifying bit errors caused by signal attenuation or noise. The receiver detects errors and signals retransmission or drops the frame.
Error Correction The receiver identifies and corrects bit error(s) *without* requesting retransmission.
EDC Error Detection & Correction bits added to a frame.
Parity Checking
- **Single Bit Parity:** Adds one bit to detect single-bit errors.
- **Two-Dimensional Parity:** Arranges data in a grid; computes parity for rows and columns. Can detect and correct single-bit errors.

Internet Checksum Used to detect errors in a transmitted segment (unreliable).
- **Sender:** Treats contents (e.g., UDP) as a sequence of 16-bit integers and computes the addition (sum) of the segment content.
- **Receiver:** Computes the checksum of the received segment; checks if the computed value equals the field value.

CRC (Cyclic Redundancy Check) A powerful error-detection technique used in Ethernet/Wi-Fi. Treats data as a polynomial; uses a generator **G** to compute a remainder. If error detected, frame is dropped.

6.3 Multiple Access Links and Protocols

Shared Medium A channel where multiple nodes can transmit, creating the possibility of collisions (e.g., Wireless, old Bus Ethernet).
Collision When two or more nodes transmit simultaneously, signals interfere.
Channel Partitioning Protocols Divide resources into dedicated pieces.
- **TDMA:** Fixed time slots.
- **FDMA:** Fixed frequency bands.

Random Access Protocols Nodes transmit at will; protocol recovers from collisions.
- **Slotted ALOHA:** Nodes transmit at start of next slot.
- **CSMA (Carrier Sense Multiple Access):** "Listen before you talk."
- **CSMA/CD (Collision Detection):** "Listen while you talk." Used in Bus Ethernet. If collision detected, stop transmitting.

Taking Turns Protocols Polling or Token Passing to coordinate access.

6.4 Switched Local Area Networks (LANs)

MAC Address A 48-bit "physical" address allocated by IEEE. Manufacturers buy address space chunks. It is "flat" and portable (moves with the NIC).
ARP (Address Resolution Protocol) Translates IP address to MAC address.
- **ARP Table:** Maps IP to MAC. Entries have a **TTL** (typically 20 min) after which mappings are forgotten.

- **ARP in Action:**
  1. A wants to send to B. B's MAC not in A's table.
  2. A broadcasts ARP Query (Dest MAC = `FF-FF-FF-FF-FF-FF`) containing B's IP. All LAN nodes receive it.
  3. B replies to A with ARP Response (unicast) giving its MAC address.
  4. A receives reply, adds B to ARP table.

Routing to Another Subnet Sending from A to B via Router R.
- A creates IP datagram (Src: A, Dest: B).
- A encapsulates datagram in frame (Dest MAC: **R's MAC**).
- R receives frame, extracts datagram, passes to IP.
- R determines outgoing interface, creates new frame (Dest MAC: **B's MAC**).

Ethernet The "dominant" wired LAN technology. Simpler, cheaper, high speed (10Mbps - 1600Gbps).
- **Topology:** Bus (old, collision domain); Switched (modern, active Layer 2 switch, no collisions).
- **Frame Structure:**
  - **Preamble:** 7 bytes of `10101010` + 1 byte of `10101011`. Synchronizes receiver clocks.
  - **Addresses:** 6-byte Dest/Src MAC. Adapter accepts frame if Dest matches its own or is broadcast.
  - **Type:** Indicates upper-layer protocol (e.g., IP) for demuxing.
  - **CRC:** Cyclic redundancy check. Frame dropped if error detected.
- **Connectionless/Unreliable:** No handshaking; no ACKs/NAKs.

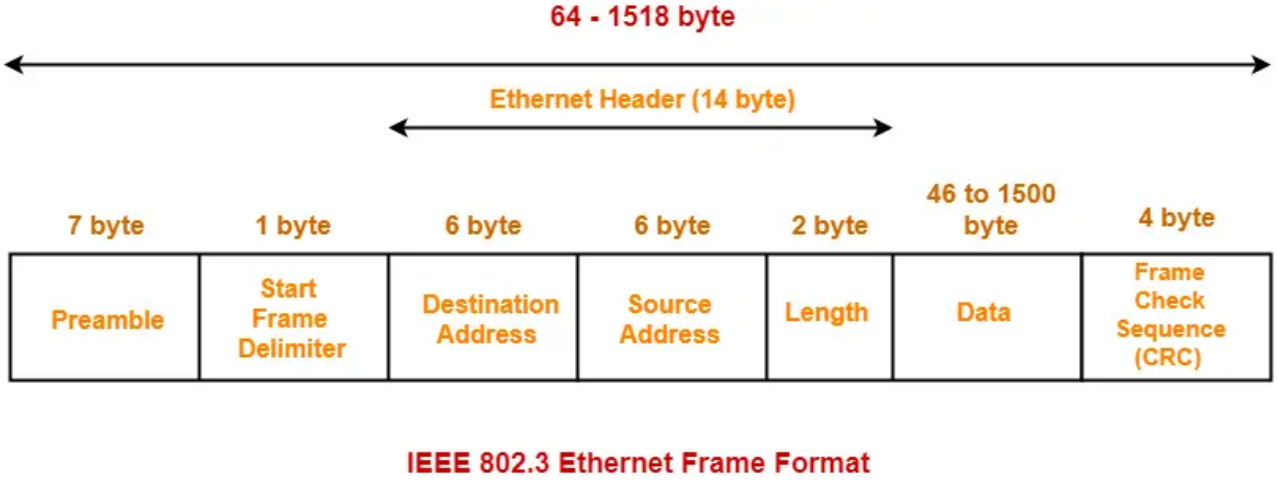
Link-Layer Switch Active Layer 2 device. Store-and-forward.
- **Function:** Examines incoming frame's MAC, selectively forwards to one or more links. Uses CSMA/CD to access segment.
- **Transparent:** Hosts are unaware of switch.
- **Plug-and-Play:** Self-learning; no config needed.
- **Simultaneous Transmission:** A-to-A' and B-to-B' can transmit at same time without collision.

Switch Forwarding/Filtering 1. Record incoming link/MAC of sender (learning).
2. Index switch table using destination MAC.
3. **If entry found:** If dest on same segment, drop frame. Else, forward to indicated interface.
4. **Else (not found):** Flood (forward on all interfaces except arriving one).

Switch vs. Router Both store-and-forward. Routers use IP/Routing Algorithms. Switches use MAC/Learning/Flooding.

6.5 Synthesis: A Day in the Life of a Web Request

Steps for a mobile client connecting to network and requesting `www.google.com`.
1. **DHCP (Connect):** Laptop needs IP, Router IP, DNS IP. Encapsulates DHCP in UDP/IP/Ethernet. Broadcasts. Router (DHCP server) replies with IP/DNS info.
2. **ARP (Before DNS):** Needs MAC of first-hop router to send DNS query. Broadcasts ARP query. Router replies with MAC.
3. **DNS:** Client sends DNS query (in UDP/IP/Eth) to Router. Router forwards to DNS server (via RIP/OSPF/BGP). DNS replies with Google's IP.
4. **TCP:** Client opens TCP socket to Google IP. Sends SYN. Web server sends SYNACK. Connection Established.
5. **HTTP:** Client sends HTTP Request into TCP socket. IP Datagram routed to Google. Google sends HTTP Reply.



IEEE 802.3 Ethernet Frame Format